

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Shaik Fahad (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:



Apply Backtracking Search to find a valid assignment.

Given

Time windows:-

$w_1 = 9-11 \text{ AM}$

$w_2 = 11 \text{ AM} - 1 \text{ PM}$

$w_3 = 1-3 \text{ PM}$

Constraints:-

- 1) Truck 1 cannot be assigned to w_3
- 2) Truck 2 must be scheduled before Truck 3.
- 3) Only one truck is assigned to each window.
- 4) Truck 3 cannot be assigned to w_1 .
- 5) Every Truck must be assigned exactly one time window.

Backtracking steps:-

- 1) Assign Truck 1 $\rightarrow w_1$

$T_1 \rightarrow w_1$

2) Assign Truck 2 $\rightarrow$ W2

$T_1 \rightarrow W_1$

$T_2 \rightarrow W_2$

3) Assign Truck 3 $\rightarrow$ W3

$T_1 \rightarrow W_1$

$T_2 \rightarrow W_2$

$T_3 \rightarrow W_3$

Constraint Check:-

1) Truck 1 not in 1-3 PM.

2) Truck 2 before Truck 3.

3) One Truck per time window.

4) Truck 3 not in 9-11 AM

5) Every Truck assigned once.

Result:-

Truck 1 $\rightarrow$ 9-11 AM

Truck 2 $\rightarrow$ 11 AM - 1 PM

Truck 3 $\rightarrow$ 1-3 PM

$\therefore$ All constraints are satisfied.

BFS Search Algorithm:-

Given constraints:-

- * 5×5 city-blocks grid
- * Start (S) to Goal (G)
- * Allowed moves:- up, down, left, right
- * Some streets are one way
- * Cost of each move = 1
- * Cannot move through blocked or restricted cells.
- * Manhattan Distance heuristic

BFS Procedure:-

- 1) Start from node S
- 2) Insert S into the queue
- 3) Mark S as visited
- 4) Remove the front node from the queue.
- 5) Expand all valid neighbouring nodes.

6) Ignore blocked cells and one-way restricted moves.

7) Add unvisited neighbours to the queue

8) Repeat untill (Goal G_1) is reached.

9) Trace back the shortest path.

Node Expansion & Frontier updates.

Step	Expanded Node	Frontier.
1.	S	Adjacent valid nodes
2.	First Neighbour	Remaining neighbours.
3.	Next Neighbour	Updated frontier
4.	Continue level by level	Queue updated
5.	Final G_1	Goal Reached.

Manhattan Distance Heuristic

$$h(n) = |x_{goal} - x_n| + |y_{goal} - y_n|$$

Result:-

The BFS Search algorithm finds the shortest path from start (S) to Goal (G_1) while satisfying given movement constraints.